**Common Photovoltaic Governing Equations**

The fundamental photovoltaic equations remain **identical** for both Se-P and TCT configurations, as the interconnection scheme does not alter the intrinsic physics of the PV cell.

**2.1 Voltage–Temperature Relation**

$$V = V_{oc,stc}(1 + \beta(T_{cell} - T_{stc}))$$

where

- $V_{oc,stc}$is the open-circuit voltage at standard test conditions,
- $\beta$is the voltage temperature coefficient,
- $T_{cell}$is the operating cell temperature.

**2.2 Current–Irradiance Relation**

$$I = I_{sc,stc}\left(\frac{G}{G_{stc}}\right)(1 + \alpha(T_{cell} - T_{stc}))$$

where

- $I_{sc,stc}$is the short-circuit current at STC,
- $G$is the incident irradiance,
- $\alpha$is the current temperature coefficient.

**2.3 Power Output**

$$P = V \times I$$

**2.4 Array-Level Aggregation**

$$V_{array} = n_{series} \times V_{module}$$
$$I_{array} = n_{parallel} \times I_{string}$$

**Important:**
These equations are **common to both Se-P and TCT configurations**.
The interconnection scheme does not modify the PV device equations.

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### 3. Series–Parallel (Se-P) Configuration

#### 3.1 Description

In the Se-P configuration, PV modules are connected in series to form strings, and multiple strings are connected in parallel to achieve the desired voltage and current levels.

#### 3.2 Electrical Behavior

- The **string current is limited by the weakest module**.
- Partial shading, temperature gradients, or orientation differences significantly reduce array output.
- High mismatch losses occur under non-uniform conditions.

#### 3.3 Wiring Characteristics

- Simple wiring layout
- Fewer interconnections
- Lower installation complexity and cost
- Limited fault tolerance

#### 3.4 Suitability

Se-P is suitable for:

- Uniform irradiance conditions
- Small-scale systems
- Cost-constrained installations

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### 4. Total Cross-Tied (TCT) Configuration

#### 4.1 Description

The TCT configuration introduces cross-ties between adjacent rows of series-connected modules, allowing current to flow laterally across the array.

#### 4.2 Electrical Behavior

- Enhanced **current redistribution**

- Reduced mismatch losses
- Higher usable current under non-uniform conditions
- Improved power extraction during partial shading or thermal gradients

Mathematically, this improvement is represented as:

$$P_{TCT} = \eta_{mismatch} \times P_{SeP}$$

where $\eta_{mismatch} > 1$.

Typical conservative values reported in literature:

- $\eta_{mismatch} = 1.05$ to $1.08$

**4.3 Wiring Characteristics**

- Increased number of interconnections
- Slightly higher cable length
- Improved redundancy
- Lower hotspot probability
- Better fault tolerance

**4.4 Suitability**

TCT is particularly effective for:

- Rooftop PV systems
- BIPV applications
- Arrays with complex geometries (V-shape, Inverted-V)
- Environments with non-uniform irradiance and temperature

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**5. Physical and Geometrical Considerations**

**5.1 Physical Effects**

- Rooftop installations experience unavoidable:
    - Temperature gradients

    - Edge cooling effects
    - Orientation-dependent irradiance
- Se-P amplifies these non-uniformities.
- TCT mitigates them through current sharing.

### 5.2 Geometrical Compatibility

- Advanced geometries (V-shape, Inverted-V) inherently create spatial non-uniformity.
- TCT complements such geometries by electrically balancing unequal module outputs.

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### 6. Wiring-Level Perspective (Practical View)

Although explicit wire-level current flow is not simulated, the **dominant wiring effect** is captured:

| Aspect | Se-P | TCT |
|---|---|---|
| Current bottleneck | High | Low |
| Hotspot risk | Higher | Reduced |
| Cable redundancy | Low | Higher |
| Long-term reliability | Moderate | High |

Explicit wire-level simulation is unnecessary for performance comparison studies and is typically reserved for protection or fault-analysis research.

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### 7. Comparative Summary

| Criterion | Se-P | TCT |
|---|---|---|
| Governing equations | Same | Same |
| Mismatch losses | High | Low |
| Power output (real rooftop) | Lower | Higher |

| Criterion | Se-P | TCT |
|---|---|---|
| Wiring complexity | Low | Moderate |
| Thermal reliability | Moderate | High |
| Suitability for complex geometry | Poor | Excellent |

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### 8. Final Conclusion

The Series–Parallel and Total Cross-Tied configurations share identical photovoltaic governing equations, as the intrinsic behavior of PV modules remains unchanged. However, their system-level performance differs due to the manner in which current is redistributed under non-uniform operating conditions. While Se-P offers simplicity and lower wiring complexity, TCT provides superior performance by mitigating mismatch losses, enhancing thermal reliability, and improving energy yield in rooftop environments. Therefore, for modern rooftop and geometrically complex PV installations, the Total Cross-Tied configuration is the preferred interconnection scheme.

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### 10. One-Sentence Reviewer-Safe Statement

The Series–Parallel and Total Cross-Tied configurations are modeled using identical photovoltaic equations, with performance differences arising solely from mismatch mitigation and current redistribution effects at the array level.